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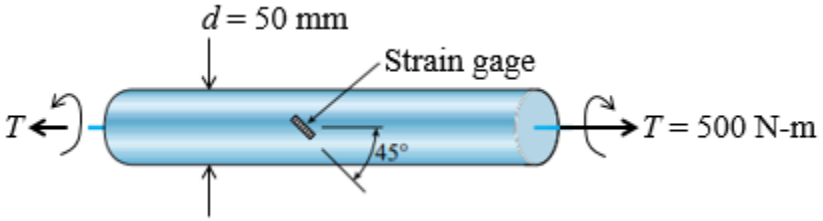
FIRST SEMESTER 2021-22

ME/MF F211: Mechanics of Solids

Comprehensive Examination: Part- A (CLOSED BOOK)

Max Marks: 30 Duration: 60 minutes December 17, 2021

Q1. A solid circular bar (see **FigQ1**) of diameter ***d*** = 50 mm is twisted in a testing machine until the applied torque reaches the value ***T*** = 500 N-m. At this value of torque, a strain gage oriented at 45° to the axis of the bar gives a reading $\varepsilon = 350 \times 10^{-6}$. What is the shear modulus (***G***) of the material? **[10]**



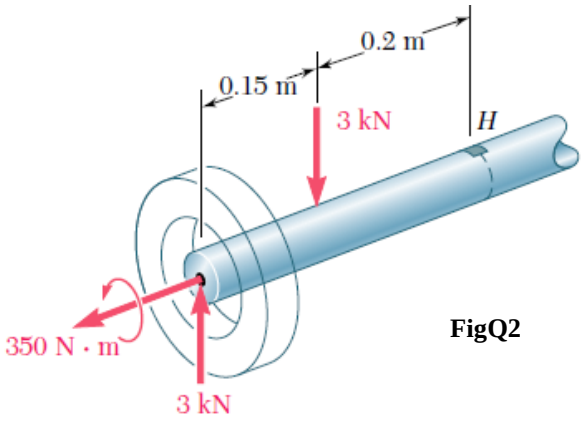
FigQ1

Q2. The axle of an automobile is acted upon by the forces and couple as shown in **Fig Q2**. Knowing that the diameter of the solid axle is 32 mm and yield strength of the material is 350 MPa. Determine: **[10]**

(a) the state of stress at point ***H*** located on top of the axle (on vertical diameter).

(b) the principal stresses at the same point.

(c) will the axle yield as per Von-Mises criteria?



FigQ2

Q3. A straight cylindrical bar ***AB*** is 15 mm diameter and 2 m long.

In first case **(i.e. Case 1):** ***AB*** is simply supported at its two ends in a horizontal position and loaded with a uniformly distributed load of intensity 60 N/m over its full span. The center deflection is found to be 5 mm.

Now in second case **(i.e. Case 2):** ***AB*** is placed vertical and loaded along its longitudinal axis.

(a) What load would cause ***AB*** to buckle in Case 2?

(b) What is the ratio of maximum bending stress in Case 1 to maximum stress in Case 2? **[10]**

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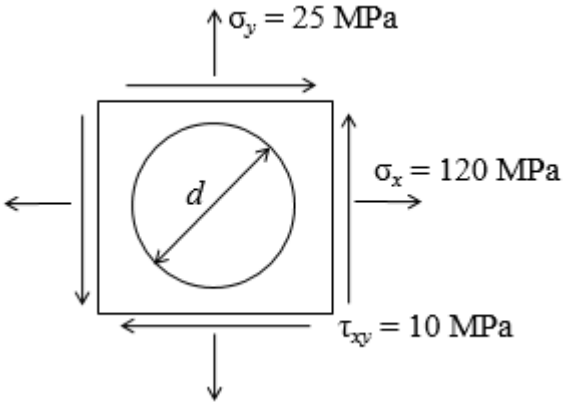
FIRST SEMESTER 2021-22

ME/MF F211: Mechanics of Solids

Comprehensive Examination: Part- B (OPEN BOOK)

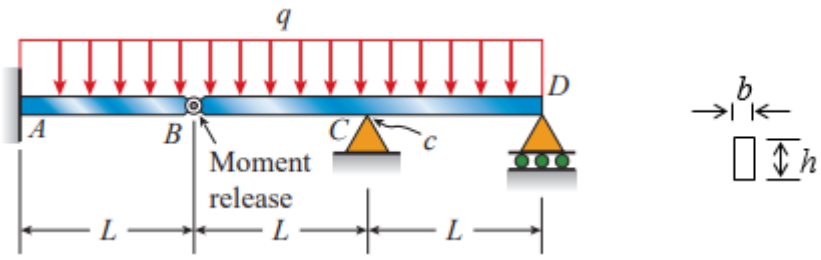
Max Marks: 50 Duration: 120 minutes December 17, 2021

Q1. A circle of diameter (***d***) 250 mm is scribed on a mild steel plate before it is subjected to the stresses as shown in **FigQ1**. In stressing, the circle deforms to an ellipse. If E = 200 GPa and ν = 0.3, calculate the lengths of the major and minor axes of the ellipse. Sketch the ellipse and show the orientation (**θ**) of major and minor axes wrt *xy* axis. **[10]**



FigQ1

Q2. The compound beam of rectangular cross-section as shown in **FigQ2** consists of a cantilever beam *AB* (Length ***L*** = 2 m), that is pin connected at *B* to a simple beam *BD* (Length **2*L*** = 4 m). After the beam is constructed, a clearance '***c* = 1 mm**' exists between the beam and a support at *C*, midway between points *B* and *D*. Subsequently, a uniform load is placed along the entire length of the beam. What intensity '***q***' of the load is needed to close the gap at *C* and bring the beam just in contact with the support? **Take**: E = 209GPa, ***h*** = 150mm, ***b*** = 100mm **[15]**



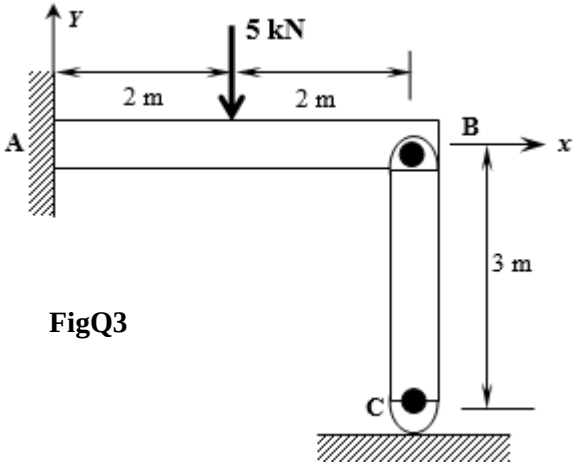
FigQ2

Q3. The beam *AB* shown in **FigQ3** is built-in to the wall at *A* and pin jointed to a 20 mm diameter column *BC* at *B*. The moment of inertia of the beam *AB* about its neutral axis, $I_{zz} = 186\times10^6$ mm^4. The joint at *C* is also a pin joint. If *E* = 210 GPa for both members

(a) Determine the deflection of the beam *AB* at point *B*.

(b) Determine the force developed in the column *BC*.

(c) Check whether the column *BC* is acceptable for the design based on the stability criteria. **[10]**



FigQ3

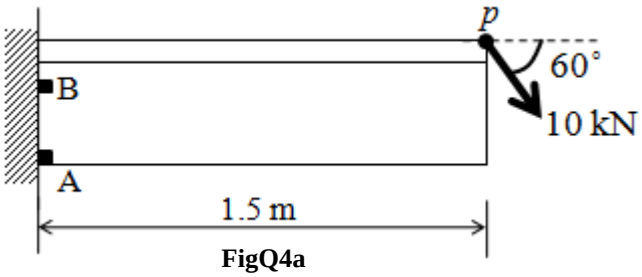
Q4. A cantilever beam of *T*-section is loaded by an inclined force of magnitude 10 kN (**FigQ4a**). The line of action of the force is inclined at an angle of 60° to the horizontal and intersects the top of the beam (i.e. point *p*) at the free end cross section. The beam is 1.5 m long and the cross section has dimensions shown in **FigQ4b**.

(a) Locate the neutral axis of the cross-section and compute the moment of inertia (i.e. I_{zz}) of the cross-sectional area about this axis.

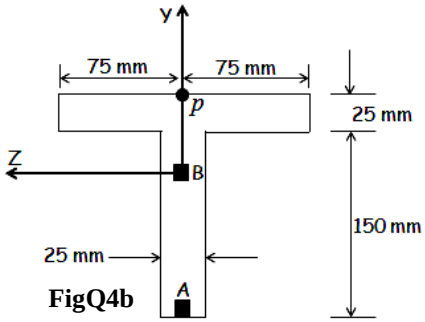
(b) Calculate the axial force *F*, shear force *V*, and bending moment M_b at a section which is very close to the fixed end of the beam (i.e. left end of the beam).

(c) Determine the stress components (i.e. σ_x and τ_{xy}) on elements at points *A* and *B* located in the web of the beam cross section. Note that element ***B*** is on neutral axis.

(d) Determine the principal stresses (i.e. σ_1 and σ_2) at point *B*. **[15]**



FigQ4a



FigQ4b